# Exact Low-Dimensional Quantum Hodology

From Qubit Action Histories to a Two-Dimensional Qutrit Phase Chart

RL Experiments Project

A pedagogical companion to the exact-construction notebooks

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**Abstract**

Can nine spatial goals be realized by a qubit or qutrit, even though neither system has nine mutually orthogonal states? The answer is yes, but the meaning of realization matters. Hilbert-space dimension bounds perfect one-shot discrimination; it does not bound the number of distinct quantum states, controlled histories, or goals. This chapter develops that distinction from first principles and gives three exactly soluble models.

First, four nonorthogonal qubit states form a projective Pauli orbit whose optimal control cost is exactly the Euclidean metric of one square cell. Second, two rationally independent qubit phase rotations give a faithful action of $\mathbb{Z}^2$; nine nonorthogonal orbit goals have the exact open $3 \times 3$ Manhattan word metric, and unit-cost retry instruments give exact Euclidean distance in expectation. This construction is algebraically exact but physically fragile because the orbit is dense on a one-dimensional Bloch circle. Third, a qutrit supplies two independent relative phases. Carefully chosen commuting generators have Fubini–Study covariance $\frac{3}{16}I_2$; an order-11 orbit contains 121 separated states, and nine of them form an exact rank-two Euclidean control-cost chart.

Along the way we prove the one-shot orthogonality obstruction, finite-automaton universality, the Bellman criterion for exact hitting costs, Schoenberg's Euclidean-distance criterion, and a general projective-orbit retry theorem. Numerical experiments validate every closed-form result and expose the recognition, tolerance, topology, and memory-provenance limitations. The result is both an introduction to the mathematics and a precise foundation for asking when hodological distance can become ordinary physical space.

**Keywords:** quantum instruments; reinforcement learning; stochastic shortest path; goal geometry; hodological space; Euclidean distance matrix; qubit; qutrit; Fubini–Study metric

## Contents

# 1 The question, stated carefully

The guiding idea is operational. For an agent, a goal that can be achieved by a short and reliable sequence of interventions is near; a goal that requires many uncertain interventions is far. The resulting space of goal-directed difficulty is a *hodological space*. It need not initially resemble physical space. It may be directed, asymmetric, high-dimensional, or partly stored in the agent's memory.

The project asks whether a special regime can occur in which:

1. many place-like goals share a common control repertoire;
2. optimal all-pairs difficulty behaves like distance;
3. the distance matrix embeds in two or three Euclidean dimensions;
4. action sequences become trajectories through recovered coordinates;
5. the geometry is carried by the controlled physical process rather than merely written into a classical goal counter.

Earlier constructions used nine orthogonal basis states of a nine-dimensional Hilbert space as nine places. Such a model makes localization simple, but it raises a natural challenge: is dimension nine essential, or can a qubit or qutrit support the same goal geometry using nonorthogonal states and richer sequence goals?

There are two superficially contradictory facts:

- A $d$-dimensional Hilbert space contains continuously many pure states. A qubit is not limited to two possible states.
- No more than $d$ nonzero states can be identified perfectly by one common, error-free, one-shot measurement.

The entire subject lies in understanding which fact is relevant to a particular meaning of place. This document separates several meanings, proves exact possibility and impossibility results, and then uses simulation to audit the gap between algebraic exactness and operational robustness.

## 1.1 A map of the argument

Section 2 reviews quantum states, instruments, and measurement backaction. Section 3 separates physical, predictive, and goal-progress state. Section 4 introduces stochastic-shortest-path control and section 5 gives the exact Euclidean embedding test. Section 6 proves what low Hilbert dimension does and does not forbid. Section 7 then converts group metrics into exactly soluble quantum control problems.

The constructions follow in increasing strength. The qubit Pauli square in section 8 is the minimal nondegenerate cell. The irrational-phase qubit in section 9 scales to the requested $3 \times 3$ patch but is fragile. The qutrit phase model in section 11 supplies a genuine two-dimensional state manifold and a finite separated orbit. Section 13 presents the numerical checks, while section 15 organizes the emerging necessary-and-sufficient conditions.

# 2 Quantum preliminaries

## 2.1 States and pure-state rays

A finite quantum system is described externally by a Hilbert space $\mathcal{H} \simeq \mathbb{C}^d$. A density operator is a positive semidefinite, unit-trace matrix

$$\rho \succeq 0, \qquad \operatorname{Tr} \rho = 1. \tag{1}$$

Pure states have rank one and may be written $\rho = |\psi\rangle\langle\psi|$ with $\langle\psi \mid \psi\rangle = 1$. Multiplication of $|\psi\rangle$ by a global phase does not change $\rho$:

$$|e^{i\gamma}\psi\rangle\langle e^{i\gamma}\psi| = |\psi\rangle\langle\psi|. \tag{2}$$

The physical pure state is therefore a *ray*, not a particular vector. This is why projective rather than ordinary unitary representations naturally appear below.

The real affine dimension of the density-operator space is $d^2 - 1$. A qubit has a three-dimensional Bloch ball of states,

$$\rho = \frac{1}{2}(I + \boldsymbol{r} \cdot \boldsymbol{\sigma}), \qquad \|\boldsymbol{r}\|_2 \leq 1, \tag{3}$$

and pure states lie on its two-dimensional boundary sphere. Thus Hilbert-space dimension is not a count of possible states.

## 2.2 POVMs and quantum instruments

A measurement outcome is represented by a positive operator-valued measure (POVM) $\{E_o\}$:

$$E_o \succeq 0, \qquad \sum_o E_o = I, \qquad \mathbb{P}(o \mid \rho) = \operatorname{Tr}(E_o \rho). \tag{4}$$

A POVM specifies probabilities but not the post-measurement state. An *instrument* specifies both. For action $a$ and reported outcome $o$,

$$\mathcal{E}_o^a(\rho) = \sum_k K_{o,k}^a \rho K_{o,k}^{a\dagger}, \qquad \sum_{o,k} K_{o,k}^{a\dagger} K_{o,k}^a = I. \tag{5}$$

The outcome probability and conditional update are

$$p(o \mid \rho, a) = \operatorname{Tr} \mathcal{E}_o^a(\rho), \qquad \rho' = \frac{\mathcal{E}_o^a(\rho)}{p(o \mid \rho, a)}. \tag{6}$$

The completeness relation is the first check for every Kraus construction in this chapter [6, 9].

A unitary action is a one-outcome instrument with $K = U$ and $U^\dagger U = I$. A random-unitary retry instrument has two outcomes,

$$K_{\mathrm{s}} = \sqrt{p}\, U, \qquad K_{\mathrm{f}} = \sqrt{1-p}\, I. \tag{7}$$

It is valid because the Kraus Gram sum is $pI + (1-p)I = I$. Its special importance is operational: success applies $U$, failure leaves the conditional state unchanged, and the agent observes which occurred.

### 2.3 Physical distance is not control distance

For later interpretation, distinguish fidelity, trace distance, and Fubini–Study distance:

$$F(\rho,\sigma) = \left(\operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right)^2, \tag{8}$$

$$D_{\mathrm{tr}}(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1, \tag{9}$$

$$d_{\mathrm{FS}}(\psi,\phi) = \arccos|\langle\psi \mid \phi\rangle|. \tag{10}$$

For pure states, $F = |\langle\psi \mid \phi\rangle|^2$ and $D_{\mathrm{tr}} = \sqrt{1-F}$. These quantify physical overlap or statistical distinguishability. They need not agree with *control cost*. A central lesson below is that four physical states can form a regular tetrahedron under state overlap while forming a square under optimal action cost [7, 8].

# 3 Three state spaces and where geometry can hide

### 3.1 Physical, predictive, and goal-progress state

The physical state is the density operator $\rho_t \in \mathcal{D}(\mathcal{H})$. The agent in an operational reinforcement-learning problem need not be given $\rho_t$; it may be a hidden state used only by the simulator.

The agent observes a history

$$h_t = (a_0, o_0, \ldots, a_{t-1}, o_{t-1}). \tag{11}$$

Two histories are predictively equivalent when no future controlled experiment can distinguish them:

$$h \sim_{\mathrm{pred}} h' \iff p(\boldsymbol{o} \mid h, \boldsymbol{a}) = p(\boldsymbol{o} \mid h', \boldsymbol{a}) \quad \text{for every future } (\boldsymbol{a}, \boldsymbol{o}). \tag{12}$$

An equivalence class is an operational causal or predictive state. In a known Markovian quantum model, the posterior density operator is sufficient, but a learner may instead compress history into a recurrent representation.

A sequence goal is a language over action–outcome symbols. A regular language is recognized by a deterministic finite automaton (DFA),

$$q_{t+1} = \delta(q_t, a_t, o_t), \qquad q_t \in Q, \tag{13}$$

with accepting set $F_g \subseteq Q$. The Markov state of the goal-conditioned problem is the product

$$(\rho_t, q_t) \in \mathcal{D}(\mathcal{H}) \times Q. \tag{14}$$

The automaton size $|Q|$ is not bounded by $d$. A recurrent network can store similar progress continuously, but changing the representation does not remove the information requirement.

*Remark* 3.1 (The provenance question). When an experiment displays a two-dimensional embedding, ask what would remain if the quantum instrument were replaced by an identity channel while the goal automaton were left intact. Geometry that survives unchanged is geometry of external task memory, not evidence that the quantum process carries space.

# 4 Hodological cost as stochastic shortest path

## 4.1 From goal pursuit to expected cost

Let $S$ be a finite operational state set. At state $i$, action $a$ costs $c(i,a) > 0$ and produces state $j$ with probability $P_a(i,j)$. For goal $g$, define the first hitting time

$$\tau_g = \inf\{t \geq 0 : S_t = g\} \tag{15}$$

and the policy cost

$$D^\pi(i,g) = \mathbb{E}_\pi\left[\sum_{t=0}^{\tau_g-1} c(S_t, A_t) \,\middle|\, S_0 = i\right]. \tag{16}$$

The hodological distance candidate is

$$D(i,g) = \inf_\pi D^\pi(i,g). \tag{17}$$

This turns easy and hard into expected intervention units. It also makes clear that distance is a property of states, actions, costs, outcomes, and goal semantics together.

## 4.2 Deriving the Bellman equation

Suppose $i \neq g$ and the first selected action is $a$. The agent immediately pays $c(i,a)$, reaches $j$ with probability $P_a(i,j)$, and then faces the same optimization problem. Conditioning on that first transition gives

$$Q_g(i,a) = c(i,a) + \sum_j P_a(i,j) D(j,g). \tag{18}$$

Choosing the least expensive first action yields

$$D(g,g) = 0, \tag{19}$$

$$D(i,g) = \min_a \left[c(i,a) + \sum_j P_a(i,j) D(j,g)\right], \qquad i \neq g. \tag{20}$$

For a proper stochastic-shortest-path problem, each goal is reached almost surely under at least one policy and improper policies have infinite or inferior cost. Under these conditions the optimal column $D(\cdot, g)$ is the minimal nonnegative solution of equation (20) [1–3].

**Proposition 4.1** (Bellman realizability)**.** *A proposed finite hitting-cost matrix is dynamically realizable by a specified proper control problem if and only if each goal column satisfies equations (19) and (20) and a minimizing proper policy exists.*

*Proof.* Necessity follows by conditioning an optimal policy on its first action. For sufficiency, select a minimizing action in each nonterminal state. Repeated substitution expresses the proposed value as expected accumulated cost plus expected residual value. Properness makes the residual vanish at the hitting time, so the selected policy attains the proposed value. Every alternative first action has value no smaller by the Bellman inequality, and induction extends this to every policy. □

### 4.3 Why hitting cost need not be a distance

In a general controlled process, $D(i,j)$ can differ from $D(j,i)$. Information gathering can be source dependent; irreversible actions can make one direction cheaper; risk penalties can violate additivity. Ordinary metric geometry requires

$$\begin{aligned} &D(i,j) = D(j,i), \qquad D(i,i) = 0, \\ &D(i,j) > 0 \quad (i \neq j), \\ &D(i,k) \leq D(i,j) + D(j,k). \end{aligned} \tag{21}$$

Failure of these conditions is not necessarily an error. It says that the appropriate geometry is directed, information-augmented, or nonmetric.

## 5 When is a finite cost matrix exactly Euclidean?

### 5.1 The double-centering construction

Suppose $D = [D_{ij}]$ is a symmetric $N \times N$ distance matrix. If unknown coordinates $x_i \in \mathbb{R}^k$ exist, then

$$D_{ij}^2 = \|x_i\|^2 + \|x_j\|^2 - 2x_i^\top x_j. \tag{22}$$

Translation does not affect distances, so assume $\sum_i x_i = 0$. Define

$$J = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top. \tag{23}$$

Multiplying the squared-distance matrix on both sides by $J$ removes the two norm terms in equation (22), leaving

$$B = -\frac{1}{2}J(D \circ D)J. \tag{24}$$

**Theorem 5.1** (Schoenberg's finite Euclidean criterion)**.** *A symmetric hollow matrix $D$ is the distance matrix of points in $\mathbb{R}^k$ if and only if*

$$B \succeq 0, \qquad \operatorname{rank} B \leq k, \tag{25}$$

*where $B$ is given by equation (24) [4, 5].*

*Proof.* If coordinates exist, the derivation above gives $B = XX^\top$, which is positive semidefinite and has rank at most $k$. Conversely, if $B = V\Lambda V^\top$ has nonnegative eigenvalues and rank $r \leq k$, set $X = V_r\Lambda_r^{1/2}$. Then $B = XX^\top$. Reversing equation (22) recovers the original distances, so the rows of $X$ are the required coordinates. □

### 5.2 Classical multidimensional scaling

The constructive half of the theorem is classical multidimensional scaling (MDS). Positive eigenvalues of $B$ give coordinates. Negative eigenvalues signal that the proposed distances are not exactly Euclidean. More than $k$ positive eigenvalues means exact embedding needs more than $k$ dimensions.

For approximate data, normalized raw stress is

$$\operatorname{stress}(X) = \sqrt{\frac{\sum_{i<j}(\|x_i - x_j\| - D_{ij})^2}{\sum_{i<j} D_{ij}^2}}. \tag{26}$$

Low stress is useful evidence; zero stress together with equation (25) is an exact certificate.

**Corollary 5.2** (Finite exact classification). *For a specified finite operational control problem, exact Euclidean hodological geometry in $\mathbb{R}^k$ is equivalent to:*

1. *proper Bellman realizability;*
2. *symmetry and the metric axioms; and*
3. *a positive-semidefinite Schoenberg Gram matrix of rank at most $k$.*

This classifies the cost matrix. It does not prove that actions are local, that the agent can localize itself, or that the geometry survives noise.

# 6 What low dimension forbids—and what it does not

## 6.1 Finite-automaton universality

**Proposition 6.1** (Automaton universality). *Let $G = (V, E)$ be any finite directed graph with positive edge costs. If goal recognition may use an unrestricted finite automaton, a physical Hilbert space of dimension $d = 1$ can realize the shortest-path hodological geometry of $G$ exactly.*

*Proof.* Create one automaton state $q_v$ per vertex. For each edge $e = (v, w)$ provide an action $a_e$ that changes $q_v$ to $q_w$ when legal and assign it the edge cost. Let every physical action be the one-dimensional identity channel with one deterministic outcome. Goal $g_w$ accepts at $q_w$. The Bellman equation of the controller is exactly the shortest-path recursion on $G$, while the physical system does nothing. □

Unit action cost does not rescue the claim. An integer-cost edge can be replaced by a chain of auxiliary automaton states, and rational costs can be implemented by retry processes. Therefore a plotted goal grid is not by itself evidence of a physical spatial structure.

## 6.2 Perfect one-shot recognition

**Proposition 6.2** (Orthogonality obstruction). *Suppose states $\rho_1, \ldots, \rho_N$ must be identified without error by one common POVM $\{E_i\}_{i=1}^N$:*

$$\mathrm{Tr}(E_i \rho_j) = \delta_{ij}. \tag{27}$$

*Their supports are mutually orthogonal. Consequently $N \leq d$.*

*Proof.* For $i \neq j$, positivity and $\mathrm{Tr}(E_i \rho_j) = 0$ imply that $E_i$ vanishes on the support of $\rho_j$. Meanwhile, $\mathrm{Tr}(E_j \rho_j) = 1$ and $0 \preceq E_j \preceq I$ imply that $E_j$ acts as the identity there. The support of $\rho_j$ is thus contained in the unit eigenspace of $E_j$ and in the kernel of every $E_i$ with $i \neq j$. Distinct supports are orthogonal, and their dimensions sum to at most $d$. □

For pure states the proof is especially transparent: $E_i|\psi_i\rangle = |\psi_i\rangle$ and $E_i|\psi_j\rangle = 0$ imply

$$\langle \psi_j \mid \psi_i \rangle = \langle \psi_j | E_i | \psi_i \rangle = 0. \tag{28}$$

An adaptive finite measurement tree acting on one copy is still one POVM when all branches are collected, so adaptivity does not evade the bound. Repeated fresh preparations or persistent history change the premise.

### 6.3 Reversibility, commuting directions, and boundaries

A channel with a completely positive trace-preserving inverse on the whole matrix algebra is unitary. Exact reversible memoryless translations must therefore be unitary channels. A maximal connected commuting subgroup of projective qubit unitaries has one phase direction. A qutrit has two independent relative phases, so $d = 3$ is minimal for a smooth two-dimensional commuting phase manifold of the kind built in section 11.

This does *not* imply that a qubit cannot encode $\mathbb{Z}^2$. A countable faithful subgroup can be dense in its phase circle; that exceptional construction is section 9. It is algebraically two-coordinate but not a robust two-dimensional manifold.

A reversible action on a finite set is a permutation, so repeated homogeneous translation produces cycles rather than an open boundary. An exact finite open grid must break boundary homogeneity, use irreversible boundary actions, store boundary position elsewhere, or appear as a patch of a larger periodic or continuous space. The qutrit construction uses the last option.

### 6.4 Why local lattice actions do not give exact Euclidean cost

Four unit-cost axial moves yield Manhattan distance. Adding unit-cost diagonals yields Chebyshev distance. Pricing a unit diagonal at $\sqrt{2}$ gives the octile metric, which is exact at offsets $(1, 0)$ and $(1, 1)$ but not at $(2, 1)$:

$$1 + \sqrt{2} \neq \sqrt{5}. \tag{29}$$

An exact finite Euclidean all-pairs metric therefore needs a sufficiently rich action repertoire, a continuous control limit with a Riemannian action cost, or a different definition of difficulty.

## 7 The projective-orbit retry theorem

The next theorem is the common engine behind the exact Euclidean examples.

**Theorem 7.1** (Projective-orbit retry construction)**.** *Let $G$ be a finite or countable group with a left-invariant metric $d_G$, rescaled so that $\min_{h\neq e} d_G(e, h) = 1$. Suppose $G$ has a projective unitary representation $U : G \to PU(d)$ and a fiducial density operator $\rho_0$ with distinct orbit labels:*

$$U_g \rho_0 U_g^\dagger = U_h \rho_0 U_h^\dagger \quad \Longrightarrow \quad g = h. \tag{30}$$

*For each $h \neq e$, define*

$$K_{\mathrm{s}}^{(h)} = \sqrt{p_h}\, U_h, \qquad K_{\mathrm{f}}^{(h)} = \sqrt{1 - p_h}\, I, \qquad p_h = \frac{1}{d_G(e, h)}. \tag{31}$$

*If goal $g$ means that the accumulated successful group product equals $g$, then the optimal expected number of unit-cost interventions from $x$ to $g$ is exactly*

$$D(x, g) = d_G(x, g). \tag{32}$$

*Proof.* The instrument is valid because

$$K_{\mathrm{s}}^{(h)\dagger} K_{\mathrm{s}}^{(h)} + K_{\mathrm{f}}^{(h)\dagger} K_{\mathrm{f}}^{(h)} = p_h I + (1 - p_h) I = I. \tag{33}$$

Failure leaves the group coordinate at $x$; success changes it to $hx$. Retry the direct displacement $h = gx^{-1}$. The number of attempts is geometric with mean

$$\mathbb{E} T_h = \frac{1}{p_h} = d_G(e, h) = d_G(x, g), \tag{34}$$

so optimal cost is no larger than the metric distance.

For the lower bound define $V(x) = d_G(x, g)$. The one-step Bellman expression for any $h$ is

$$1 + (1 - p_h)V(x) + p_h V(hx). \tag{35}$$

The triangle inequality and left invariance imply

$$V(x) - V(hx) \le d_G(x, hx) = d_G(e, h) = \frac{1}{p_h}. \tag{36}$$

Rearranging shows that the Bellman expression is at least $V(x)$. No action, adaptive policy, or mixture can do better, while direct retry attains $V$. $\square$

*Remark* 7.2 (What the theorem does and does not explain). The theorem proves exact optimality, but it places the desired metric in the success probabilities. It is an analytic benchmark, not yet a derivation of the Euclidean norm from simpler physics. The deeper inverse-design question is why a natural instrument family should produce $p_h = 1/d_G(e, h)$ rather than some other law.

# 8 The minimal exact cell: a qubit Pauli square

## 8.1 Four nonorthogonal physical states

Let $\mathcal{H} = \mathbb{C}^2$ and choose a pure fiducial state with Bloch vector

$$\boldsymbol{r}_{00} = \frac{1}{\sqrt{3}}(1, 1, 1). \tag{37}$$

Let $X, Y, Z$ denote the Pauli matrices. Although $XZ = -ZX$, their conjugation channels commute because the minus sign is a global phase:

$$XZ\rho ZX = ZX\rho XZ. \tag{38}$$

The orbit under $X$ and $Z$ has Bloch vectors

$$\begin{aligned} \boldsymbol{r}_{00} &= (+,+,+)/\sqrt{3}, \qquad & \boldsymbol{r}_{10} &= (+,-,-)/\sqrt{3}, \\ \boldsymbol{r}_{01} &= (-,-,+)/\sqrt{3}, \qquad & \boldsymbol{r}_{11} &= (-,+,-)/\sqrt{3}. \end{aligned} \tag{39}$$

These are the vertices of a regular tetrahedron. For distinct labels,

$$\mathrm{Tr}(\rho_{uv}\rho_{u'v'}) = \frac{1 + \boldsymbol{r}_{uv} \cdot \boldsymbol{r}_{u'v'}}{2} = \frac{1}{3}. \tag{40}$$

The physical states are nonorthogonal and cannot be four perfectly distinguishable qubit places.

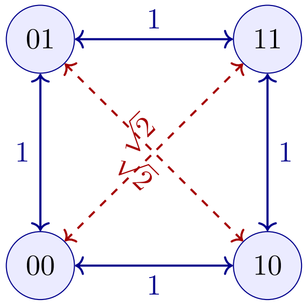
## 8.2 Actions and exact cost

Use deterministic axial channels

$$\mathcal{X}(\rho) = X\rho X, \qquad \mathcal{Z}(\rho) = Z\rho Z \tag{41}$$

and a stochastic diagonal instrument

$$K_{\mathrm{s}} = 2^{-1/4}Y, \qquad K_{\mathrm{f}} = \sqrt{1 - \frac{1}{\sqrt{2}}}\, I. \tag{42}$$




Physical overlap: tetrahedral
Optimal cost: square Euclidean
Hilbert dimension: 2

**Figure 1:** The smallest nondegenerate example. State geometry is tetrahedral, but the action-cost geometry is exactly the Euclidean unit square.

A successful $X$, $Z$, or $Y \simeq XZ$ toggles the history parity by $(1,0)$, $(0,1)$, or $(1,1)$ modulo two. The goal monitor has four states $q = (u,v) \in \mathbb{Z}_2^2$.

Ordering the goals as $00, 10, 01, 11$, optimal movement cost is

$$D = \begin{pmatrix} 0 & 1 & 1 & \sqrt{2} \\ 1 & 0 & \sqrt{2} & 1 \\ 1 & \sqrt{2} & 0 & 1 \\ \sqrt{2} & 1 & 1 & 0 \end{pmatrix}. \tag{43}$$

Axial neighbors require one deterministic action. Opposite corners can be reached by two axial actions at cost two or by retrying the diagonal instrument, whose mean cost is $\sqrt{2}$. The latter is optimal by theorem 7.1.

The Schoenberg matrix of equation (43) is positive semidefinite of rank two and reconstructs $(0,0), (1,0), (0,1), (1,1)$. This is the cleanest demonstration that hodological geometry need not reproduce ambient quantum state geometry.

### 8.3 Minimality and limitation

Four points are the smallest repertoire containing a complete square cell. A one-dimensional quantum system can imitate the square only by storing all four states in an external automaton. A qubit is the smallest nontrivial system with the projective Pauli $\mathbb{Z}_2^2$ orbit. The construction is exact, but it closes after two steps in either direction and therefore does not scale to an open $3 \times 3$ lattice.

## 9 An exact open qubit lattice from irrational phases

### 9.1 Definition of the orbit

Take

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad U = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\beta} \end{pmatrix} \tag{44}$$

with

$$\alpha = \frac{2\pi}{9} + \epsilon\sqrt{2}, \qquad \beta = \frac{2\pi}{3} + \epsilon\sqrt{3}, \qquad \epsilon = 10^{-2}. \tag{45}$$

The four cardinal actions are the single-Kraus instruments $U, U^\dagger, V, V^\dagger$. Define nine goals

$$G_{ij}: \quad |\psi_{ij}\rangle = U^i V^j |+\rangle, \qquad i,j \in \{0,1,2\}. \tag{46}$$

These are nine distinct, nonorthogonal phases on the qubit equator.

## 9.2 Orbit–stabilizer viewpoint

Let an action group or semigroup $\Gamma$ act on a fiducial ray. Its projective stabilizer is

$$K_\psi = \{k \in \Gamma : \pi(k)|\psi_0\rangle \sim |\psi_0\rangle\}. \tag{47}$$

The orbit is naturally labelled by the quotient $\Gamma/K_\psi$. If action words have norm $\|\cdot\|_{\mathcal{A}}$, the induced cost of displacement $\Delta$ is

$$d_{\mathrm{hod}}(0, \Delta) = \min_{k \in K_\psi} \|\Delta + k\|_{\mathcal{A}}. \tag{48}$$

Faithfulness, $K_\psi = \{0\}$, preserves the whole word metric. On a finite patch, trivial stabilizer is stronger than necessary: it is enough that no stabilizer relation shortens any displacement in the patch difference set.

**Theorem 9.1** (Faithfulness of the irrational qubit orbit)**.** *The map*

$$(a, b) \longmapsto U^a V^b |+\rangle \tag{49}$$

*with equation (45) is injective on rays for* $(a, b) \in \mathbb{Z}^2$.

*Proof.* Suppose $U^a V^b|+\rangle \sim |+\rangle$. The coefficient of $|0\rangle$ is unchanged, so the global phase must be one. Equality of the relative phase requires

$$a\alpha + b\beta = 2\pi m, \qquad a, b, m \in \mathbb{Z}. \tag{50}$$

Substituting equation (45),

$$2\pi\left(\frac{a}{9} + \frac{b}{3} - m\right) + \epsilon(a\sqrt{2} + b\sqrt{3}) = 0. \tag{51}$$

The second term is algebraic. If the rational coefficient of $\pi$ were nonzero, the equation would express $\pi$ as an algebraic number, contradicting its transcendence. Therefore

$$\frac{a}{9} + \frac{b}{3} - m = 0, \qquad a\sqrt{2} + b\sqrt{3} = 0. \tag{52}$$

Squaring the second relation gives $2a^2 = 3b^2$. Unique prime factorization forces $a = b = 0$; then $m = 0$. The stabilizer is trivial. □

## 9.3 Exact Manhattan word metric

Because $U$ and $V$ commute, every cardinal action word reduces to $U^a V^b$. To move $G_{ij}$ to $G_{kl}$, faithfulness forces

$$(a, b) = (k - i, l - j). \tag{53}$$

Any word realizing this displacement contains at least $|a|$ horizontal and $|b|$ vertical generators. The canonical word attains the bound, so

$$\boxed{d_1(G_{ij}, G_{kl}) = |k - i| + |l - j|}. \tag{54}$$

This is exactly the open $3 \times 3$ square-lattice graph metric, even though all nine rays lie on a single Bloch circle.

### 9.4 Sequence-language formulation

For a fixed horizon $H$, goal $G_{ab}$ can be stated as the finite language of action sequences whose signed $U$ and $V$ exponent totals equal $(a, b)$. A DFA may track

$$(n_U, n_V, t) \in [-H, H]^2 \times \{0, \ldots, H\}, \tag{55}$$

with a dead state for illegal or overlong histories. Taking $H = 4$ retains every shortest transition among the nine goals.

Without a horizon, exact equality of two unbounded integer counters is not a regular language. Calling it regular would silently introduce an infinite memory. The physical orbit remains mathematically well defined, but a finite agent needs either bounded tasks, approximate counters, or a different recognition mechanism.

### 9.5 Exact Euclidean distance with unit-cost instruments

Cardinal actions give $L^1$ distance. For each nonzero patch displacement $\delta = (a, b)$, define

$$r_\delta = \sqrt{a^2 + b^2}, \qquad K_{\mathrm{s}}(\delta) = \sqrt{\frac{1}{r_\delta}}\, U^a V^b, \qquad K_{\mathrm{f}}(\delta) = \sqrt{1 - \frac{1}{r_\delta}}\, I. \tag{56}$$

Every attempt costs one. Since nonzero integer displacements have $r_\delta \geq 1$, this is a valid instrument. Failure changes nothing; success performs the exact displacement.

**Corollary 9.2** (Exact qubit Euclidean patch)**.** *For the nine goals in equation (46), optimal expected cost under equation (56) is*

$$\boxed{d_2(G_{ij}, G_{kl}) = \sqrt{(k-i)^2 + (l-j)^2}}. \tag{57}$$

*Proof.* The direct displacement has geometric waiting time of mean $r_\delta$. Any composite strategy with successful displacements $\delta_1, \ldots, \delta_n$ has expected cost at least

$$\sum_q \|\delta_q\|_2 \geq \left\| \sum_q \delta_q \right\|_2 \tag{58}$$

by the Euclidean triangle inequality. The direct retry action is optimal. This is also the specialization of theorem 7.1 to $\mathbb{Z}^2$. □

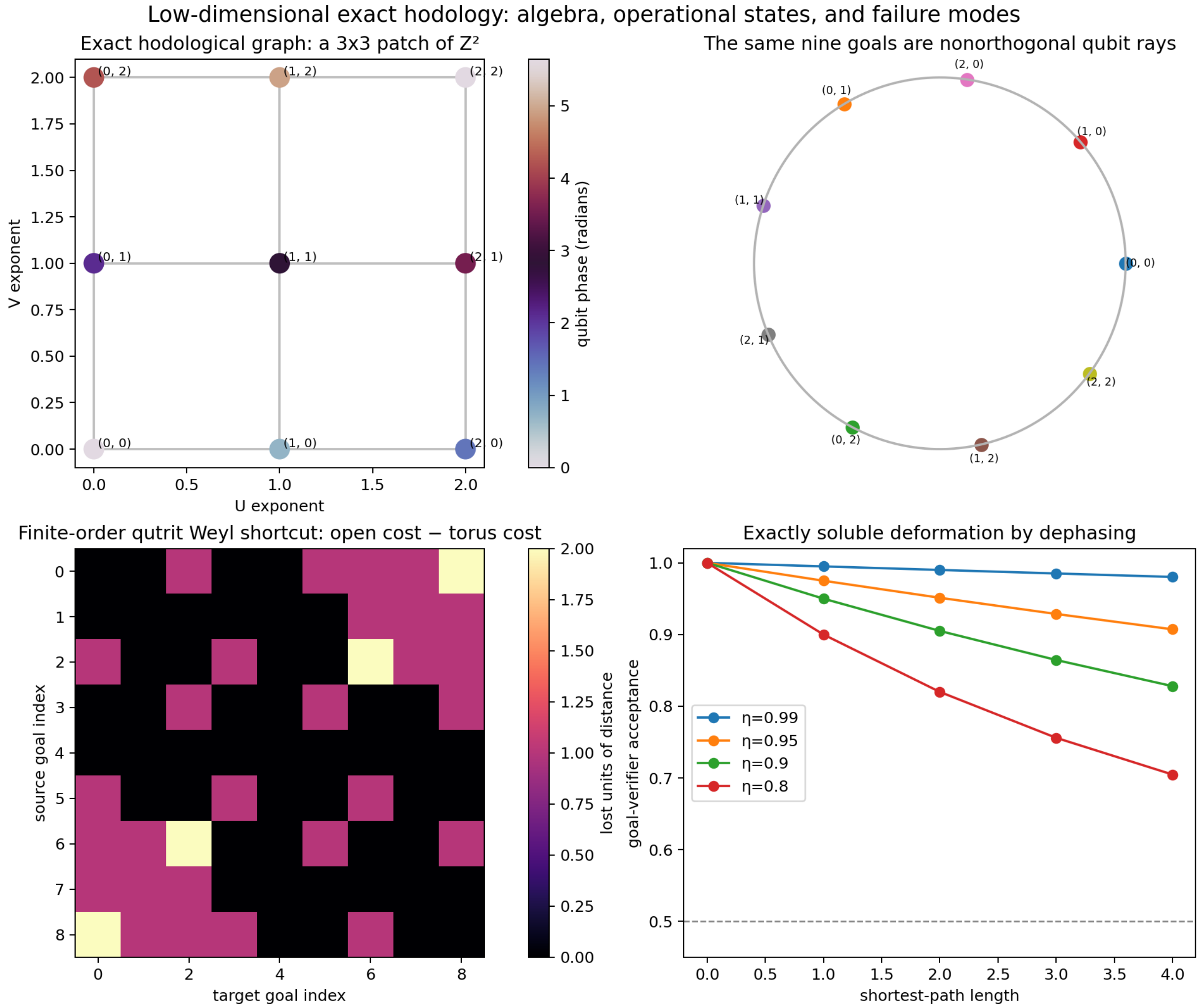



**Figure 2: Exact qubit construction and its failure modes.** The upper panels show the open action-cost lattice and the nine physical rays on one equatorial circle. The lower panels show finite-order qutrit wraparound and the exactly soluble dephasing response.

# 10 Recognition and robustness of the qubit construction

## 10.1 One-sided verification is not exclusive localization

Goal $G_{ij}$ has the yes/no projective test

$$M_{\text{yes}} = P_{ij} = |\psi_{ij}\rangle\langle\psi_{ij}|, \qquad M_{\text{no}} = I - P_{ij}. \tag{59}$$

It accepts the exact target with probability one. A different goal is falsely accepted with probability

$$\mathbb{P}(\text{yes} \mid G_{kl}) = |\langle\psi_{ij} \mid \psi_{kl}\rangle|^2 = \cos^2\left(\frac{(i-k)\alpha + (j-l)\beta}{2}\right). \tag{60}$$

For the chosen phases, minimum angular separation is 0.6352 radians and worst false acceptance is 0.9025. Independent repreparations reduce a one-sided error $q$ to $q^n$, but no finite number perfectly discriminates nonorthogonal states.

Thus the exact theorem concerns history-labelled equality and control cost. An experimental realization must include the resource needed to track history or statistically recognize the target.

**Table 1:** Finite-tolerance shortcuts for the exact qubit construction.

| Allowed infidelity | shortened pairs | exact mean cost | tolerant mean cost |
|---:|---:|---:|---:|
| $10^{-12}$ | 0.0000 | 2.0000 | 2.0000 |
| $10^{-5}$ | 0.0000 | 2.0000 | 2.0000 |
| $10^{-3}$ | 0.3056 | 2.0000 | 1.6111 |
| $10^{-2}$ | 0.3611 | 2.0000 | 1.5000 |

## 10.2 Dense orbit and approximate stabilizers

The irrational subgroup is dense on the equatorial phase circle. Hence arbitrarily distant integer labels can produce arbitrarily close physical rays. The inverse map from ray to $(a,b)$ is discontinuous. A useful finite-resolution diagnostic is

$$\ell_\varepsilon = \min\left\{|a|+|b| :\ (a,b)\neq(0,0),\ 1-|\langle +|U^aV^b|+\rangle|^2 \le \varepsilon\right\}. \tag{61}$$

Small $\ell_\varepsilon$ means that an approximate stabilizer creates a cheap shortcut.

Enumerating exponent pairs through radius 18 gives: Exact injectivity therefore does not imply robust localization. Compactness eventually defeats a large, noise-tolerant encoding without redundancy.

## 10.3 Exactly soluble homogeneous dephasing

After each intended unitary, apply

$$\mathcal{D}_\eta(\rho) = \frac{1+\eta}{2}\rho + \frac{1-\eta}{2} Z\rho Z. \tag{62}$$

This channel multiplies qubit off-diagonal coherence by $\eta$. A path of length $L$ multiplies it by $\eta^L$. The target projector therefore accepts with probability

$$\boxed{p_{\text{accept}}(L,\eta) = \frac{1+\eta^L}{2}}. \tag{63}$$

The noise is homogeneous and depends only on path length, so reliability decays radially while the word geometry remains isotropic. Direction-dependent dephasing after $U$ and $V$ would instead produce anisotropic, Finsler-like difficulty.

*Proof.* The equatorial state has density matrix

$$\rho_\varphi = \frac{1}{2}\begin{pmatrix} 1 & e^{-i\varphi} \\ e^{i\varphi} & 1 \end{pmatrix}. \tag{64}$$

One application of equation (62) multiplies the off-diagonal entries by $\eta$; induction gives $\eta^L$. Overlap with the ideal target ray of the same phase is the trace of their product, $(1+\eta^L)/2$. □

# 11 A robust two-dimensional qutrit phase chart

## 11.1 Why a qutrit changes the problem

A normalized qutrit pure state has two independent relative phases. Diagonal unitaries modulo global phase form a two-torus, so two commuting control directions can span a genuine two-dimensional state manifold. We now choose the fiducial amplitudes and generators so that its local metric is exactly isotropic.

Let

$$|\psi_0\rangle = \sqrt{\frac{3}{8}}|0\rangle + \frac{1}{2}|1\rangle + \sqrt{\frac{3}{8}}|2\rangle \tag{65}$$

and define

$$A = \mathrm{diag}(0, 1, 0), \qquad B = \mathrm{diag}\left(0, \frac{1}{2}, 1\right). \tag{66}$$

They commute. The continuous orbit is

$$|\psi(\theta, \phi)\rangle = e^{i(\theta A + \phi B)}|\psi_0\rangle. \tag{67}$$

## 11.2 Deriving the Fubini–Study metric

For a normalized pure-state family, the infinitesimal Fubini–Study line element is

$$ds^2 = \langle d\psi \mid d\psi\rangle - |\langle\psi \mid d\psi\rangle|^2. \tag{68}$$

Differentiating equation (67),

$$|d\psi\rangle = i(A\, d\theta + B\, d\phi)|\psi\rangle. \tag{69}$$

Substitution into equation (68) gives the covariance form

$$ds^2 = \mathrm{Var}(A)d\theta^2 + 2\,\mathrm{Cov}(A, B)d\theta\, d\phi + \mathrm{Var}(B)d\phi^2. \tag{70}$$

Because $A$ and $B$ commute with the orbit unitaries, these moments are constant everywhere on the orbit.

The fiducial probabilities in the computational basis are $(3/8, 1/4, 3/8)$. Therefore

$$\begin{aligned}
\langle A\rangle &= \frac{1}{4}, & \langle A^2\rangle &= \frac{1}{4}, & \mathrm{Var}(A) &= \frac{1}{4} - \frac{1}{16} = \frac{3}{16},\\
\langle B\rangle &= \frac{1}{8} + \frac{3}{8} = \frac{1}{2}, & \langle B^2\rangle &= \frac{1}{16} + \frac{3}{8} = \frac{7}{16}, & \mathrm{Var}(B) &= \frac{7}{16} - \frac{1}{4} = \frac{3}{16},\\
\langle AB\rangle &= \frac{1}{8}, & \mathrm{Cov}(A, B) &= \frac{1}{8} - \frac{1}{4}\frac{1}{2} = 0.
\end{aligned} \tag{71}$$

Hence

$$\boxed{ds^2 = \frac{3}{16}(d\theta^2 + d\phi^2)}. \tag{72}$$

The manifold is locally flat and square in these phase coordinates, up to one common scale. A generic fiducial and pair of diagonal generators would produce unequal variances or a cross term; the cancellation here is deliberate and exact.

### 11.3 A finite faithful orbit

Choose an odd integer $m \geq 5$ and set

$$\epsilon_m = \frac{4\pi}{m}, \qquad U = e^{i\epsilon_m A}, \qquad V = e^{i\epsilon_m B}. \tag{73}$$

For the experiments, $m = 11$. Both $U$ and $V$ have order $m$ and commute.

**Proposition 11.1** (Faithfulness for odd order)**.** *For odd $m$, the projective orbit*

$$(x, y) \longmapsto U^x V^y |\psi_0\rangle, \qquad (x, y) \in \mathbb{Z}_m^2, \tag{74}$$

*has $m^2$ distinct rays.*

*Proof.* All three fiducial amplitudes are nonzero. If $U^x V^y$ stabilizes the ray, the $|0\rangle$ component fixes the global phase to one. The phases on $|1\rangle$ and $|2\rangle$ require

$$2x + y \equiv 0 \pmod{m}, \qquad 2y \equiv 0 \pmod{m}. \tag{75}$$

Because $m$ is odd, two is invertible modulo $m$. Thus $y = 0$, then $x = 0$. $\square$

For $m = 11$ the qutrit has 121 distinct, nonorthogonal orbit states. Their number can exceed $d$ because they are not required to be one-shot distinguishable.

### 11.4 The square-torus control metric

For a residue $r \in \mathbb{Z}_m$, let $\bar{r}$ be its centered representative in

$$\left\{-\frac{m-1}{2}, \ldots, \frac{m-1}{2}\right\}. \tag{76}$$

Define

$$d_m((x, y), (x', y')) = \sqrt{\bar{r}^2 + \bar{s}^2}, \quad r = x' - x, \quad s = y' - y. \tag{77}$$

This is a left-invariant control metric on the finite group $\mathbb{Z}_m^2$. Its unit step is the common scale factored out of equation (72).

For each nonzero centered displacement $(r, s)$, let

$$W_{rs} = U^r V^s, \qquad L_{rs} = \sqrt{r^2 + s^2}, \qquad p_{rs} = \frac{1}{L_{rs}}, \tag{78}$$

and use the instrument

$$K_{\mathrm{s}}^{(r,s)} = \sqrt{p_{rs}}\, W_{rs}, \qquad K_{\mathrm{f}}^{(r,s)} = \sqrt{1 - p_{rs}}\, I_3. \tag{79}$$

Theorem 7.1 proves that optimal expected cost is exactly $d_m$.

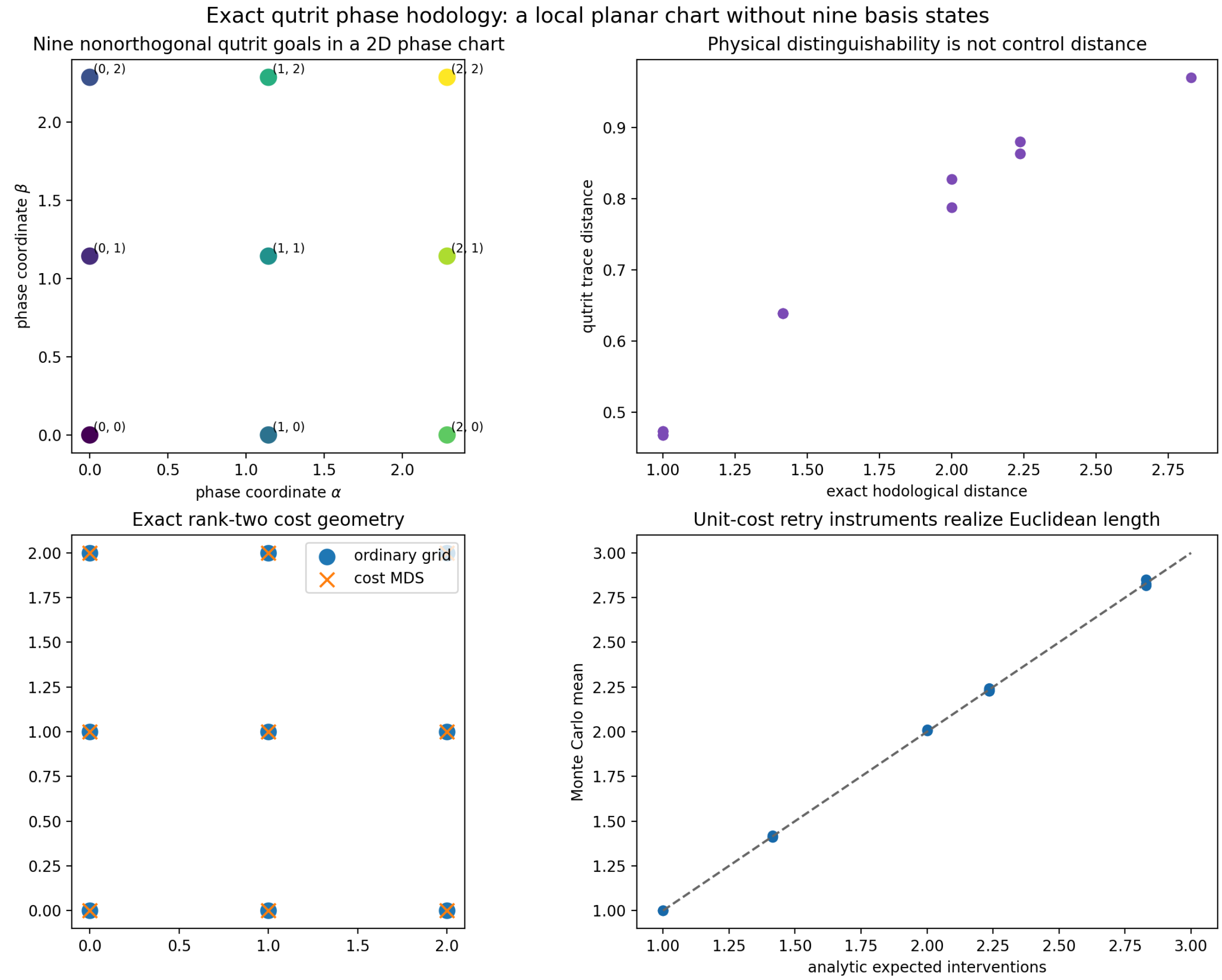



**Figure 3: Exact qutrit phase construction.** The generator covariance is isotropic; the finite order-11 orbit contains 121 separated rays; the selected patch passes the exact Euclidean-distance and Bellman tests; and Procrustes-aligned MDS coordinates coincide with the analytic grid.

## 11.5 The exact open 3-by-3 chart

Select

$$\mathcal{G}_{3\times 3} = \{(x, y) : x, y \in \{0, 1, 2\}\} \tag{80}$$

inside the $m = 11$ orbit. Pairwise coordinate differences have magnitude at most two, well below the wrap radius 5. Therefore centered residues equal ordinary differences, and

$$\boxed{D((x, y), (x', y')) = \sqrt{(x - x')^2 + (y - y')^2}}. \tag{81}$$

This is the exact Euclidean distance matrix of a square lattice. Its Schoenberg Gram matrix has two positive eigenvalues, both six, and all other eigenvalues vanish to numerical precision.

There is no physical wall around the patch. It is a local chart of a larger periodic orbit. That is advantageous here: open-grid distances are obtained without an orthogonal nine-state position register or a boundary-dependent transition rule.

### 11.6 Goal semantics and dead reckoning

The clean exact monitor tracks successful displacement

$$q_t = (x_t, y_t) \in \mathbb{Z}_m^2. \tag{82}$$

With known initialization and no hidden disturbance,

$$q_t = (x, y) \quad \Longrightarrow \quad \rho_t = \rho_{xy}. \tag{83}$$

Thus physical state and history monitor evolve in lockstep. This proves existence, but it also allows dead reckoning: the agent can navigate from its own observed success outcomes without inferring location from a separate quantum probe.

The next operational test should hide the initial coordinate, insert unobserved slips, and provide only weak nonorthogonal measurements. Then the agent must infer a posterior over phase coordinates. Because the target states are nonorthogonal, localization becomes a genuine Bayesian quantum filtering problem while equation (81) remains exact ground truth.

## 12 Why the obvious qutrit Weyl construction is a torus

Define the generalized Pauli operators

$$X|j\rangle = |j + 1 \bmod 3\rangle, \qquad Z|j\rangle = \omega^j|j\rangle, \qquad \omega = e^{2\pi i/3}. \tag{84}$$

They satisfy $ZX = \omega XZ$, so their conjugation channels commute and their projective classes generate $\mathbb{Z}_3^2$. A generic fiducial has nine orbit states. This decisively refutes the claim that nine controlled states require $d = 9$.

It does not produce an open grid. Since $X^3 = Z^3 = I$, the stabilizer kernel is $3\mathbb{Z} \times 3\mathbb{Z}$. For example, open-grid displacement $(2, 0)$ has length two, but on the torus it equals $(-1, 0)$ and takes one inverse action. In the experiment, 16 of the 36 unordered goal pairs have strict shortcuts.

This is an algebraic fact, not a learning failure. Finite-order relations quotient the intended translation lattice. The torus has two independent translation directions and is intrinsically two-dimensional topologically, but its nine-point shortest-path matrix is not an exact rank-two Euclidean distance matrix. Open plane, periodic topology, and Euclidean metric are different claims.

## 13 Simulation results and exact audits

### 13.1 Why simulate an exactly solved model?

An exact construction still benefits from computation. Numerical checks can:

1. catch implementation errors in Kraus operators and phase conventions;
2. verify all source–goal pairs rather than only representative cases;
3. test Monte Carlo convergence against analytic expectations;
4. quantify finite-resolution behavior not visible in an exact proof;
5. generate artifacts against which later learning agents can be tested.

The simulation does not discover the theorem. It tries to falsify the code that claims to instantiate it.

Table 2: Headline checks for the order-11 qutrit phase construction.

| Quantity | Result |
|---|---|
| Hilbert dimension | 3 |
| Physical orbit states | 121 |
| Selected goal states | 9 |
| Fubini–Study covariance | $(3/16)I_2$ |
| Minimum orbit Frobenius separation | 0.3450505 |
| Maximum Bellman residual | $8.88 \times 10^{-16}$ |
| Maximum patch Euclidean error | 0 |
| Maximum MDS reconstruction error | $1.33 \times 10^{-15}$ |
| Positive Schoenberg eigenvalues | 6, 6 |
| Control/trace-distance Pearson correlation | 0.988921 |
| Maximum waiting-time Monte Carlo error | 0.023733 |

## 13.2 Exact qubit audit

All 81 ordered transitions among the nine qubit goals reach the analytic target. The maximum floating-point terminal infidelity is $6.7 \times 10^{-16}$. The Euclidean coordinate-distance error is zero. For each of the 24 nonzero patch displacements, 50,000 geometric waiting times were sampled, for 1.2 million samples in total. Maximum error of a sample mean from the exact norm was 0.02897 interventions.

The same audit exposes the limitation: worst one-shot false acceptance is 0.9025, and at infidelity tolerance $10^{-3}$, 30.56% of ordered nontrivial pairs are shortened. The exact label geometry is much stronger than the finite-resolution physical geometry.

## 13.3 Exact qutrit audit

The control/trace correlation is high because this is a small isotropic chart, but the two metrics are conceptually different. Control cost is set by instruments and expected attempts. Trace distance is set by statistical distinguishability of physical states. Exact equality is neither assumed nor observed.

## 13.4 A dialectical reading of the evidence

The results answer a sequence of progressively sharper objections:

**Objection: nine goals require nine dimensions.**
False. Both the qubit and qutrit constructions have nine distinct controlled goals.

**Objection: perhaps the grid is only a fitted picture.**
False for the cost matrices. Bellman and Schoenberg conditions prove the exact results.

**Objection: perhaps any goal automaton could do this.**
Correct in general. The identity-system control in theorem 6.1 shows why physical orbit and memory provenance must be reported.

**Objection: the qubit orbit is a physical plane.**
False. It is dense on one phase circle. Its two-dimensionality is in the faithful action/history labels.

**Objection: the qutrit Weyl orbit is the desired open plane.**
False. It wraps into a torus.

**Objection: the qutrit phase construction finally derives Euclidean distance.**
Not yet. It derives an isotropic state manifold, but the inverse-distance retry probabilities still supply the Euclidean control norm.

## 14 What exactly has emerged?

The qubit answers the original cardinality question with the smallest possible physical system. The qutrit answers the stronger manifold question. Neither alone establishes that ordinary space generically emerges from quantum control. Rather, they isolate what remains to be explained:

- why a small local action family should select the Euclidean norm;
- how location is inferred without exact dead reckoning;
- why approximate stabilizers remain expensive under finite resolution;
- how local charts scale and agree on overlaps;
- how internal goal progress should be separated from spatial position.

**Table 3:** The three exact constructions and the distinct claims they support.

| Construction | Exact achievement | Physical strength | Principal limitation |
|---|---|---|---|
| Qubit Pauli square | One Euclidean square cell | Nontrivial projective qubit orbit | Closes after two steps |
| Qubit irrational phase | Open $\mathbb{Z}^2$ word metric and exact Euclidean patch cost | Nine nonorthogonal controlled rays | Dense one-dimensional orbit; fragile recognition |
| Qutrit order-11 phase | Isotropic two-phase manifold and exact Euclidean local chart | 121 finite separated rays | Euclidean retry law and macro-actions supplied |
| Qutrit Weyl orbit | Exact homogeneous $\mathbb{Z}_3^2$ control topology | Common finite projective group action | Periodic torus, not an open Euclidean plane |

# 15 Toward necessary and sufficient conditions for space

There cannot be one useful classification theorem until the intended meaning of space is fixed. Automaton universality makes an unrestricted statement trivial. The following hierarchy separates exact finite geometry from the stronger physical claim.

## 15.1 Level I: exact Euclidean cost geometry

For a finite operational MDP, the necessary-and-sufficient conditions are already complete:

1. every goal is properly reachable and the cost columns solve the Bellman equations;
2. the all-pairs table satisfies symmetry and the metric axioms;
3. its Schoenberg matrix is positive semidefinite of rank at most two or three.

These conditions answer a precise mathematical question: whether optimal difficulty is exactly the Euclidean distance of some point configuration.

## 15.2 Level II: spatially consistent dynamics

An arbitrary teleportation table can be tuned to have Euclidean costs. A spatial interpretation additionally requires action-conditioned increments to be local and reusable. If $x_i$ are recovered coordinates, then

$$P_a(i,j) > 0 \quad \Longrightarrow \quad x_j - x_i \in S_a(i). \tag{85}$$

Away from boundaries, the distribution of increments under action $a$ should be approximately independent of absolute position. This is translation covariance. In a curved or inhomogeneous model, its controlled failure should vary smoothly and be interpretable as a position-dependent metric or connection.

## 15.3 Level III: operationally identifiable place

The coordinate must be inferable from the agent's available history to the required accuracy. Exact one-shot identification obeys theorem 6.2; sequential approximate identification does not. Useful finite tests include localization error, posterior entropy at fixed sensing budget, and discrimination performance after hidden disturbances.

### 15.4 Level IV: low external-memory provenance

The spatial coordinate should not live entirely in a hand-coded monitor. Candidate diagnostics are:

- minimal DFA size for exact goal recognition;
- predictive Hankel rank;
- recurrent-state dimension at fixed prediction error;
- performance after erasing action labels while retaining outcomes;
- performance after randomizing the unknown initial state;
- recovery after unobserved moves;
- comparison with an identity-channel or state-independent null instrument.

A useful hybrid control stores one coordinate in orthogonal qutrit basis states and one in a three-state DFA. A correct provenance analysis should identify one quantum-supported axis and one history-supported axis.

### 15.5 Level V: robust manifold and bundle structure

As goal repertoires grow, local dimension and metric tensors should converge. Charts inferred from overlapping subsets should agree up to smooth coordinate transformations. Approximate stabilizers should not create shortcuts within the operational horizon. A useful invariant is the shortest nonzero word that returns within fidelity tolerance $\varepsilon$ of its starting ray, as in equation (61).

Only after a stable spatial base is established should internal degrees of freedom be treated as fibers. If an internal state changes under transport around a closed base loop, the learned transition law defines an operational holonomy and possibly a discrete connection. The exact flat qutrit base is valuable precisely because any measured holonomy can then be attributed to the added coupling rather than uncertainty about base geometry.

## 16 The most promising next construction

### 16.1 Remove the displacement lookup table

The exact retry catalog contains one action for every displacement class. That makes the theorem transparent but places too much geometry in the action repertoire. The strongest next experiment should retain the qutrit's isotropic phase manifold while replacing the catalog by a small family of local translation-covariant instruments.

An outer optimization can vary valid Kraus operators through Choi matrices or Stinespring isometries. For each candidate environment, an inner exact planner computes all-pairs Bellman costs. A possible objective is

$$\begin{aligned}\mathcal{L} = \lambda_B\,\mathcal{L}_{\text{Bellman}} + \lambda_-\,\|B_-\|_F^2 + \lambda_r \sum_{\ell>2} \lambda_\ell(B)^2 \\ + \lambda_{\text{iso}}\,\mathcal{L}_{\text{anisotropy}} + \lambda_{\text{nl}}\,\mathcal{L}_{\text{nonlocal}} + \lambda_{\text{tol}}\,\mathcal{L}_{\text{tolerance}}.\end{aligned} \tag{86}$$

Here $B_-$ is the negative part of the Schoenberg matrix, eigenvalues beyond the first two penalize extra dimension, and the last terms discourage nonlocal action semantics and fragile approximate stabilizers.

Training and evaluation displacements must be separated. A model that merely memorizes the $3 \times 3$ table will fail at held-out radii and directions. A genuinely emergent local metric should extrapolate.

### 16.2 Derive length from continuous control energy

There is also an analytic route that avoids retry macro-actions. Consider

$$U(t) = e^{i(\theta(t)A+\phi(t)B)}. \tag{87}$$

By equation (72), Fubini–Study speed is

$$v_{\mathrm{FS}}(t) = \frac{\sqrt{3}}{4}\sqrt{\dot{\theta}(t)^2 + \dot{\phi}(t)^2}. \tag{88}$$

If action cost is integrated speed,

$$C[\theta, \phi] = \frac{\sqrt{3}}{4}\int_0^T \sqrt{\dot{\theta}^2 + \dot{\phi}^2}\, dt, \tag{89}$$

then minimizing cost gives straight phase paths and Euclidean geodesic length. Discretizing this variational principle could make the metric arise from local control energy rather than inverse-distance success probabilities.

### 16.3 Make localization genuinely predictive

After removing the lookup table, hide the initial phase coordinate and add unobserved random slips. Provide only weak qutrit measurements whose statistics vary smoothly across the orbit. A recurrent agent must then learn:

1. a posterior or predictive state from measurement history;
2. a local action model;
3. goal-conditioned values;
4. an atlas inferred from learned all-pairs costs.

The exact phase chart supplies ground truth for separating model-estimation, belief, planning, and execution error.

## 17 Reproduction

Run from the directory containing this document:

```
python -m unittest discover -s tests -v
python -m unittest discover -s search -p 'test_*.py' -v
MPLBACKEND=Agg python run_exact_experiments.py
MPLBACKEND=Agg python run_qutrit_phase_experiments.py
MPLBACKEND=Agg python search/search_low_dimensional.py
latexmk -pdf -interaction=nonstopmode \
  -halt-on-error EXACT_LOW_DIMENSIONAL_HODOLOGY.tex
```

The exact-construction tests verify Kraus completeness, all 81 canonical qubit transitions, Manhattan word distance, bounded sequence goals, Euclidean macro-actions, geometric waiting time, qutrit wraparound, dephasing, qutrit faithfulness, Fubini–Study isotropy, Bellman residuals, and Schoenberg rank. The skeptical-search tests add automaton nulls, qutrit POVMs, toroidal Bellman costs, qubit tangent patches, and quantum counter backaction.

Primary artifacts are:

**`results/all_pairs.csv`**
Every ordered exact qubit source–goal transition.

**`results/finite_tolerance.csv`**
Approximate-stabilizer shortcut audit.

**`results/dephasing_validation.csv`**
Analytic and Monte Carlo values of equation (63).

**`results/euclidean_waiting_validation.csv`**
Geometric waiting means for 1.2 million samples.

**`results/qutrit-phase/summary.json`**
Qutrit covariance, separation, Bellman, Schoenberg, and MDS checks.

# 18 Conclusion

Low Hilbert dimension does not count goals. A qubit can carry a faithful action of $\mathbb{Z}^2$ and therefore an exact open lattice word metric; a qutrit can carry a smooth, finite-resolution two-dimensional phase manifold. What low dimension does constrain is perfect simultaneous recognition and robust localization.

The exact constructions make several distinctions unavoidable. Physical state geometry is not control geometry. A two-generator topology is not automatically a Euclidean metric. A history coordinate is not automatically a physical coordinate. A zero-stress distance matrix is not automatically a local space. Bellman and Schoenberg conditions solve the finite cost-geometry problem, while action covariance, recognition, memory provenance, and robustness determine whether the result deserves a spatial interpretation.

The qutrit phase chart is the strongest present starting point. It supplies a genuine isotropic two-dimensional quantum manifold, a finite separated orbit, and an exactly solved goal geometry. The remaining challenge is no longer whether such geometry is mathematically possible. It is to explain why a small physically natural family of local quantum interventions should select a noise-stable Euclidean metric that an agent can reconstruct from experience.

# A Worked Bellman check for a retry action

Suppose the current point is $x$, the goal is $g$, and action $h$ succeeds with probability $p = 1/r$, where $r = d_G(e, h)$. Let $V(x) = d_G(x, g)$. Its one-step value is

$$\begin{aligned} Q_h(x) &= 1 + \left(1 - \frac{1}{r}\right) V(x) + \frac{1}{r} V(hx) \\ &= V(x) + 1 - \frac{V(x) - V(hx)}{r}. \end{aligned} \tag{90}$$

The triangle inequality gives $V(x) - V(hx) \le r$, hence $Q_h(x) \ge V(x)$. If $h = gx^{-1}$, then $V(hx) = 0$ and $r = V(x)$, so equality holds. This one line contains both halves of optimality: every action is bounded below by the metric, and the direct displacement attains it.

## B Worked Schoenberg check for the unit square

For points $(0,0), (1,0), (0,1), (1,1)$, use the distance matrix in equation (43). Double centering its squared entries gives

$$B = \begin{pmatrix} \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}. \tag{91}$$

Its eigenvalues are $1, 1, 0, 0$: it is positive semidefinite of rank two. The two positive eigenvectors recover a rotated and centered version of the unit square. This is why rank, not visual inspection, certifies intrinsic dimension.

## C Notation and interpretive checklist

| Symbol | Meaning |
|---|---|
| $d$ | Hilbert-space dimension |
| $\rho$ | Hidden or externally represented density operator |
| $h_t$ | Observable action–outcome history |
| $q_t$ | Explicit goal-progress or displacement monitor state |
| $D(i, g)$ | Optimal expected intervention cost |
| $D_{\mathrm{tr}}$ | Trace distance between physical states |
| $d_{\mathrm{FS}}$ | Fubini–Study distance or line element |
| $B$ | Double-centered Schoenberg Gram matrix |
| $K_\psi$ | Projective stabilizer of the fiducial ray |
| $U, V$ | Controlled phase translations |
| $K_{\mathrm{s}}, K_{\mathrm{f}}$ | Success and failure Kraus operators |

Before interpreting any low-dimensional goal embedding, ask:

1. What is the physical Hilbert dimension?
2. What classical or recurrent memory recognizes the goal?
3. Can different coordinates be distinguished by future quantum experiments?
4. Does the proposed cost solve the Bellman equations?
5. Does its Schoenberg matrix have the claimed rank and sign?
6. Are action increments local and translation covariant?
7. Does geometry survive unknown starts, hidden slips, and finite measurement tolerance?
8. Is Euclidean length derived or inserted into action probabilities?

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